

INIYAN C

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building and deploying predictive models using Python, Scikit-learn, and deep learning frameworks. Designed and deployed EduSense AI — a full-stack ML platform serving real-time inference via REST APIs — and a Mental Health Risk Analyzer using NLP-based classification pipelines. Experienced in ETL pipeline construction, feature engineering, and data analysis with Power BI. Pursuing AWS Cloud Practitioner certification. IBM Coursera certified in ML and Python for Data Science.

TECHNICAL SKILLS

ML & Deployment: Python, Scikit-learn, TensorFlow (EfficientNet CNN), Predictive Modelling, Model Evaluation, Feature Engineering, NLP, TF-IDF, SVM

Cloud & DevOps: AWS (S3, EC2, SageMaker — in progress), Flask REST APIs, WebSocket, Git, GitHub, FFmpeg

ETL & Data: ETL Pipelines, Pandas, NumPy, SQL, MySQL, Data Preprocessing, Power BI, Matplotlib, Seaborn.

Certifications: Machine Learning with Python – IBM Coursera | Python for Data Science & AI – IBM Coursera | AWS CLF-C02 (pursuing)

WORK EXPERIENCE

Machine Learning Engineer Intern — GRADTWIN

Feb 2026 – Present

- Built and evaluated supervised predictive models in Python using Scikit-learn — implementing feature engineering, preprocessing pipelines, and cross-validation to optimise model accuracy on structured datasets.
- Delivered end-to-end ML solutions by translating business requirements into model specifications, benchmarking performance metrics, and documenting results for stakeholder review.

Data Analyst Intern — Owl AI (Overload Ware Labs AI)

Sep – Oct 2025

- Constructed ETL pipelines using Python (Pandas) to ingest, clean, and transform multi-dimensional datasets, then visualised findings in Power BI dashboards to drive stakeholder decisions.
- Identified data quality issues across 20+ reports and applied preprocessing techniques that improved downstream analysis reliability and business recommendation accuracy.

AI & ML Intern — NSIC Technical Service Centre

Jan – Feb 2025

- Executed end-to-end data analysis workflows — ingestion, cleaning, EDA, and reporting — using Python and Power BI, delivering KPI dashboards for non-technical stakeholders.

PROJECTS & RESEARCH

EduSense AI — ML-Powered Educational Platform | *Python · Flask · Groq LLM · Whisper ASR · DeepFace · REST API · WebSocket*

- Built and deployed a full-stack predictive ML platform that transcribes lecture videos via Whisper ASR and serves automated quiz generation, summarisation, and student engagement inference through 20+ REST API endpoints.
- Engineered real-time inference pipeline using Flask-SocketIO and DeepFace emotion recognition (7-class classification → attention score), plus a PDF-parsing ETL pipeline feeding an AI Exam Predictor ranked by Bloom's taxonomy.
- Deployed a Cognitive Load Analyzer, 7-Day Personalised Learning Path generator, and HOD Analytics Dashboard and multi-language TTS output in 8 Indian languages.

Mental Health Sentiment & Risk Analyzer | *Python · NLP · Scikit-learn · TF-IDF · SVM · Pandas*

- Built an end-to-end NLP classification pipeline to predict mental health risk categories (anxiety, depression, suicidal ideation, stress) using TF-IDF vectorisation and a fine-tuned SVM model on real-world Reddit datasets.
- Engineered a 3-tier risk-scoring layer (Low / Moderate / High) from model confidence and sentiment polarity, then visualised class distribution and risk trends with Matplotlib and Seaborn to surface actionable clinical insights.

Brain Tumor Classification — Deep Learning | *TensorFlow · EfficientNet CNN · MRI Analysis* | *Published COACONS 2025, Certificate of Merit*

- Trained an EfficientNet CNN model (TensorFlow) to classify brain MRI scans across multiple tumour categories — awarded Certificate of Merit at the Conference on Advances in Communication Networks & System.

EDUCATION

B.Tech — Artificial Intelligence & Data Science — **Paavai College of Engineering** 2022 – 2026 | CGPA: 7.7
Higher Secondary (CBSE) — **Kurunji CBSE School**, 2022 | Score: 75%

AWARDS

- "Engineering Excellence" Award — Uniconverge Technologies | Published researcher — ICON'26 2026